

Chapter 1: Foundations of Retrieval

Retrieval-augmented generation pairs an LLM with an external index.

Documents are chunked, embedded, and stored in a vector database.

Query embeddings find the nearest chunks, which become extra context.

Pitfall: shallow chunks miss the surrounding argument; deep chunks waste tokens.

Practical rule: chunk to natural section boundaries, not fixed token counts.

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Chapter 2: Knowledge Graph Routing

Vector search answers 'similar to' but not 'related to'.

Knowledge graphs encode typed relations, allowing multi-hop reasoning.

Hybrid: vector finds candidates, graph filters and orders them by relation.

Build graphs from sources (extracted entities) and from curated notes (Brain Graph).

Refresh the graph after each ingestion; otherwise routing drifts.

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Chapter 3: Provenance and Trust

Every answer should carry a fine-grained citation.

Fine-grained means heading anchor or paragraph, not just file path.

When the model paraphrases, the citation lets a human verify.

Provenance also enables incremental refresh: only invalidate notes whose sources changed.

Pattern: store `source_hash` in the curated note frontmatter.

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Chapter 4: Operational Hygiene

Run ingestion on a schedule, not ad hoc.

Log every decision: what was promoted, what was skipped, why.

Failures route to a review queue; nothing is silently dropped.

Treat the wiki as code: git tracks every change, including the graph snapshot.

Quarterly: walk through review/needs-human and decide each entry.

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